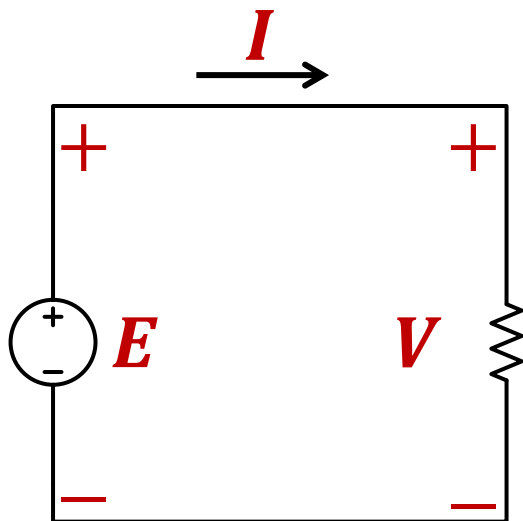




$$EI = VI$$

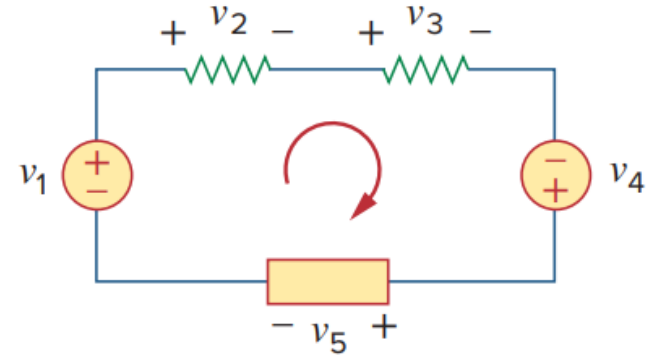
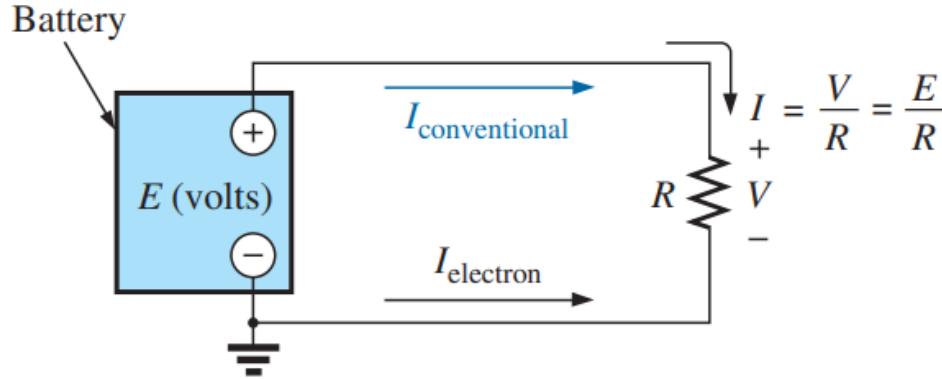
$$E = V$$

$$+E - V = 0$$

$$\sum V = 0$$

KVL: The algebraic sum of the voltages around any closed path (Loop) is zero.

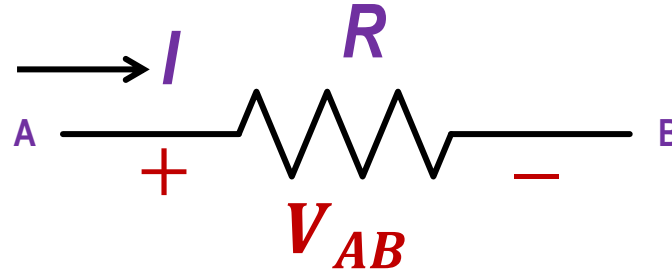
KVL: The algebraic sum of the voltages around any closed path (Loop) is zero.



Voltage rise (Low to High) = Voltage drop (High to Low)

Voltage rise (Low to High) - Voltage drop (High to Low) = 0

Between **Two Nodes**, there must be a **Voltage Difference** even if there isn't any element between them.



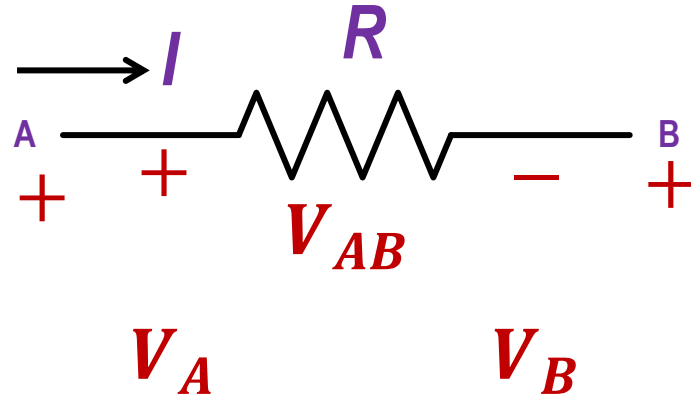
$$V_{AB} = IR$$

Voltage difference between two nodes of a resistor is equal to the voltage drop across the resistor as obtained by Ohm's law

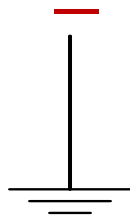
$$V_{AB} = -V_{BA}$$

NB: While applying KVL we consider voltage drop across resistors or voltage difference between two nodes.

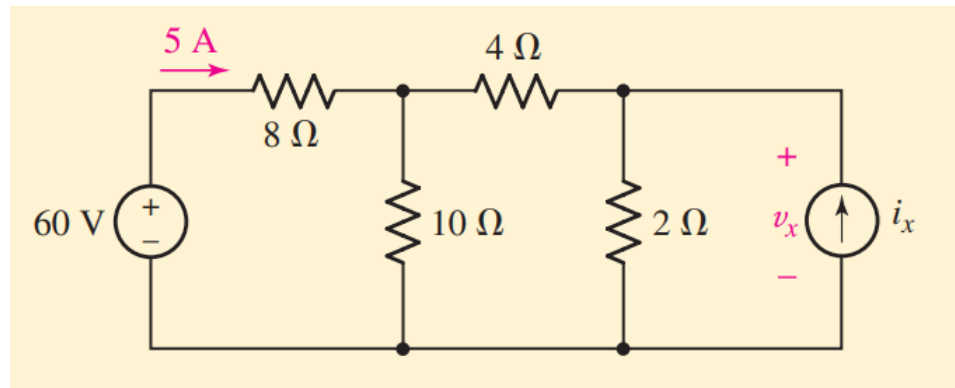
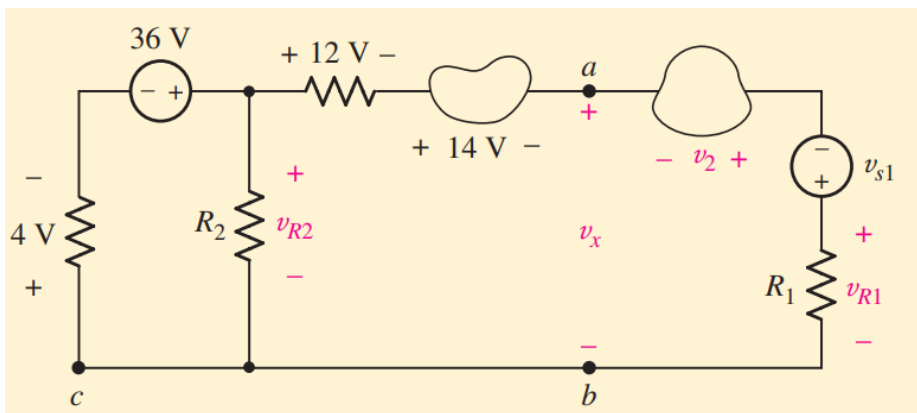
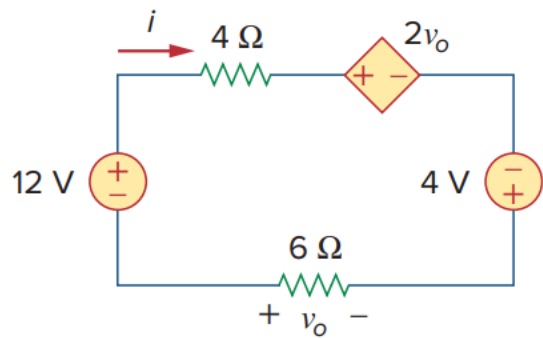
Between **Two Nodes**, there must be a **Voltage Difference**



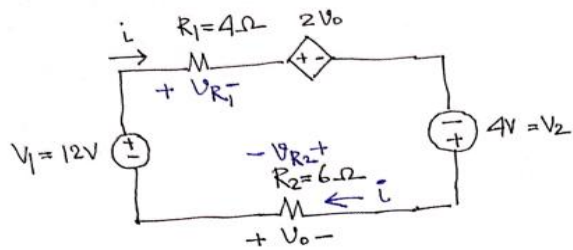
$$V_{AB} = V_A - V_B$$



Voltage between two nodes is equal to the sum of the voltage drop and rise across all the elements that exist along any path between the two nodes.



Find i & V_o of the ckt in the Fig. below.



Applying KVL

$$+V_1 - V_{R1} - 2V_o + V_2 - V_{R2} = 0$$

$$\Rightarrow 12 - 4i - 2V_o + 4 - 6i = 0$$

$$\Rightarrow -10i - 2V_o = -16$$

$$\Rightarrow 10i + 2V_o = 16 \quad \text{--- (1)}$$

From (1) $\Rightarrow$

$$10i - 2V_o = 16$$

$$\Rightarrow 10i + 2(-6i) = 16$$

$$\Rightarrow 10i - 12i = 16$$

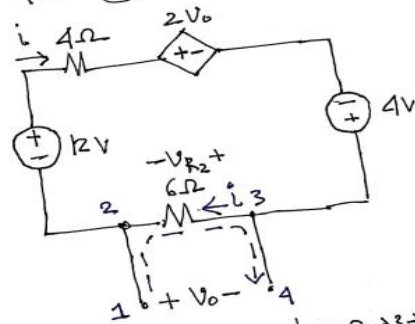
$$\Rightarrow -2i = 16$$

$$\Rightarrow i = -8A$$

$$V_o = -6i = -6 \times (-8) \\ = 48V$$

Find the Relationship between V_{R2} & V_o

Redrawing the ckt



Applying KVL around $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$

$$+V_{R2} + V_o = 0$$

$$\Rightarrow +iR_2 + V_o = 0$$

$$\Rightarrow 6i + V_o = 0$$

$$\Rightarrow V_o = -6i$$

You can apply KVL even if the circuit is not complete.

